



STREETLIGHT **INSIGHT**

StreetLight AADT 2024
(Initial Release)
Methodology and Validation
White Paper
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Version 1

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Introduction

Annual Average Daily Traffic (AADT) is the industry's fundamental metric for analyzing infrastructure projects, measuring congestion, estimating road safety, and enabling empirical measurement for approving highway funds. The StreetLight Data, Inc. ("StreetLight") InSight® platform delivers AADT for virtually any roadway on-demand, directly through the computer.

Traditionally, AADT counts are collected via permanent or temporary physical sensors, limited to a fraction of U.S. roadways, or modeled from regional demand models. Both traditional approaches result in inaccuracy. Additionally, temporary counts are cumbersome and expensive and can put staff in harm's way.

StreetLight develops an AADT estimate each year for urban and rural road segments across the U.S. that performs as well as or better than industry-standard accuracy targets and can be used by industry practitioners for traffic monitoring and reporting, traffic-impact studies, infrastructure prioritization, and more. Year after year, StreetLight's data science team continuously optimizes the AADT metric to bring the most comprehensive estimates to agencies across the U.S.

What's New in AADT 2024

AADT 2024 includes metrics powered by Aggregated GPS Data (AGPS). This includes a unique blend of device navigation apps, traditional mobile data apps, and in-vehicle navigation apps. AGPS Data has a highly representative sample with strong penetration rates across road classes or types and regions. Penetration rates are measured as sample trip counts divided by ground-truth vehicle volumes collected at permanent counter locations. On average, StreetLight's AGPS Data source captured 33% of vehicles on the road. This highly representative sample contributes to strong metrics quality.

AADT 2024 is built directly on the results of the AGPS monthly volume model. For details on the data sources and model design used in the AGPS Volume model, please see [the All Vehicles AGPS Volume white paper](#). **Estimated AADT 2024 is equal to the average of Monthly Average Daily Traffic (MADT) estimates for all 12 months in 2024, weighted by the number of days in the month.**

StreetLight AADT volume estimates include prediction intervals that represent the likely upper and lower bounds of the true value. The 95% prediction interval is an estimate of the range within which the true value is expected to lie 95% of the time. The linked white paper above includes details on the methodology.

This paper highlights the methodology and validation for our AADT 2024 initial release (in February 2025). Later in the year, we will receive data from additional permanent counters that

we will use to update this AADT model. At that time, we will update this white paper. While we understand that changing estimates might be confusing for some users, releasing initial and updated results mimics traditional reporting process for AADT. It also helps users balance the need for quick results.

Ground-Truth Data

To train and validate the AADT 2024 metric, we used vehicle volume counts from 2,688 unique permanent counter locations across 21 states in the contiguous U.S. Counters were only used if they could provide data for more than 10 months of the year.

Location of Counters in 2024 AADT

Only Includes Counters with 10+ Months of Data

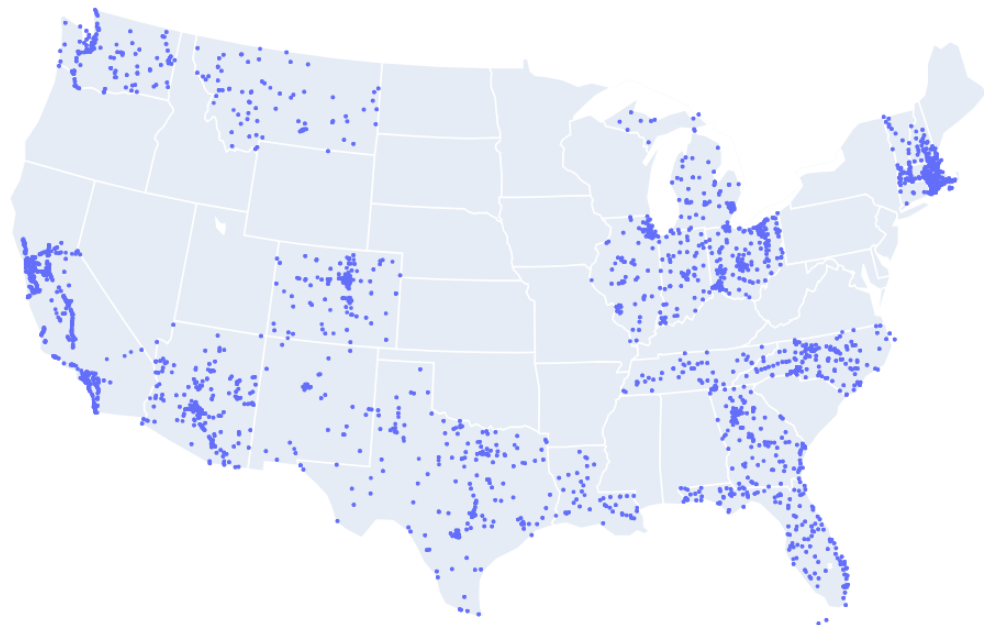


Figure 1: Location of permanent counters with data in 2024 used to train and validate the StreetLight AADT 2024 U.S. metric.

Evaluating the Accuracy of StreetLight's AADT 2024

The following sections describe accuracy definitions and present validation results for AADT 2024. To measure the accuracy of our AADT 2024 metric, we compare AADT estimates to the

ground truth data from permanent counters. Then, we evaluate the accuracy of those estimates against the industry-standard benchmark for short-term count expansion.

Defining Error Metrics

To evaluate model accuracy, we consider the following five error metrics, each of which illuminates a different aspect of model performance. Metrics are reported by road size because errors (usually measured as a percentage of permanent counter AADT) naturally increase along smaller roads. The selected error metrics align with those that are most widely utilized in the transportation industry. For technical definitions of these errors, see Krile et al (2024)¹. Smaller absolute error values indicate better accuracy for each error metric.

- **Mean absolute percentage error (MAPE)** is a standard measure of error for predictive machine-learning models.
- **Normalized root mean square error (NRMSE)** is another standard error metric in machine learning. It penalizes large errors more than MAPE, so it is more sensitive to outliers. Larger outliers lead to a larger NRMSE.
- **68th percentile absolute percentage error** represents one standard deviation from the mean within a standard bell curve and thus describes the typical error across the road segments. The 68th percentile absolute error can be interpreted as the expected typical error.
- **95th percentile absolute percentage error** represents the upper limit of the expected absolute error. It helps identify how far away more extreme errors tend to be.
- **Bias, or median percentage error**, shows whether the model tends to over or underestimate. It is less sensitive to outliers than MAPE. Values close to zero suggest that our model does not systemically under or overestimate. Positive values indicate more overestimation, and negative values indicate more underestimation.

Additionally, we consider an error metric to measure precision by road size. This measure is included in the FHWA benchmarks discussed below.

- **Maximum number of sample sites exceeding maximum error range** represents the number of counter sites within a volume category that can exceed the maximum error range set by the FHWA for that volume category. This recognizes that a limited number of sites may have large positive or negative estimation errors, but that errors should generally be low and closely distributed.

¹ Krile, R., Habtemichael, F. (2024). Guidelines for Obtaining AADT Estimates from Non-Traditional Sources (FHWA-PL-21-031). United States. Federal Highway Administration.

Sources of Error

StreetLight AADT 2024 is a sample expansion, and differences in the sample over time and space is the most common cause of error. Though we account for much of the variation in the sample with our AGPS Volume Model, we expect higher error levels in regions or on road types or classes where we have fewer (or no) permanent counters.

StreetLight relies on an algorithm that maps road segments from the segment library of our Aggregated GPS data provider to the StreetLight InSight® Zone Library, derived from OpenStreetMap (OSM). While rare, significant differences in the two segment maps are another source of error. Segment mapping issues are most common where many roads intersect, such as on busy highway interchanges or on highly circuitous segments in rural and residential areas.

StreetLight's AGPS sample size is smaller on less frequently used roads—a category which includes many rural residential roads. If sample trip counts fall below a privacy or statistical threshold, no volume estimate will be produced for this road.

Short-Term Count Expansion as an Accuracy Target

As a benchmark, we compared the accuracy of our estimates to a widely used approach: short-term count expansion. In this approach, a two-day temporary count is collected from a road segment and then expanded to estimate AADT using scaling factors from nearby permanent counters. Our aim was for StreetLight's AADT accuracy to exceed or come close to the existing short-term count expansion accuracy.

Note that no widely accepted studies have determined the accuracy of temporary count expansion for roads with fewer than 500 AADT (though temporary count expansion for these roads is frequently performed). Thus, simply knowing accuracy for such small roads is a benefit of the StreetLight approach.

Comparison to Updated FHWA Benchmarks

AADT Volume Category Range	Sample Size for Evaluation	Bias (%)	MAPE (%)	Maximum Error Range (%)
500-4,999 (low)	$N \geq 1,000$	± 1.6	13.0	± 43.4
	$N = 200$	± 3.1	14.5	
	$N = 100$	± 4.5	15.5	
	$N = 50$	± 6.4	16.7	
	$N = 25$	± 8.7	18.1	
5,000-54,999 (medium)	$N \geq 1,000$	± 2.0	10.5	± 33.6
	$N = 200$	± 2.9	11.3	

	N = 100	± 3.8	11.9	
	N = 50	± 5.2	12.7	
	N = 25	± 6.9	13.6	
55,000+ (high)	N ≥ 1,000	± 2.7	8.2	± 22.0
	N = 200	± 4.0	8.9	
	N = 100	± 5.3	9.8	
	N = 50	± 7.1	11.2	
	N = 25	± 9.4	12.9	

Table 1: Errors expected from short-term count expansion estimation of AADT serve as our accuracy targets for different volume levels and a selection of sample sizes. This is Table 14 on pg. 48 from Krile et al. (2024).¹ The table shows examples of errors at different sample sizes across three road size categories. The referenced paper includes a formula to calculate benchmarks.

Table 1 shows benchmarks for expected errors associated with a same-year, short-term count expansion for three categories of roads at different sample sizes, as estimated by the Federal Highway Administration (FHWA).² Sample sizes refer to the number of counter locations in the volume bin. These updated expected errors benchmarks are from a new report released by the FHWA in August 2024.

Table 1 includes two different types of benchmark—accuracy benchmarks (bias and MAPE) and precision benchmarks (Maximum Error Range)—for three AADT volume ranges. The accuracy benchmarks vary based on the sample size.

There are two factors to note in this table:

1. The reference targets for bias and MAPE decrease as the sample size increases. This means that the more data one has, the more accurate the estimates are expected to be.
2. Controlling for the sample size, estimates for higher volume roads are expected to be more accurate and precise as demonstrated by the smaller bias, MAPE, and Maximum Error Range.

The FHWA provides the formulas to calculate the exact accuracy benchmarks for any sample size (Table 2). We applied these formulas when comparing our performance to these benchmarks, shown later in this white paper.

AADT Volume Category	Sample Size for Evaluation	Target Bias (%)	Target MAPE (%)	Maximum Error Range (%)
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² Krile, R., Habtemichael, F. (2024). Guidelines for Obtaining AADT Estimates from Non-Traditional Sources (FHWA-PL-21-031). United States. Federal Highway Administration.

500-4,999 (low)	By Equation ($25 \leq N \leq 1,000$)	$25.65 - 15.64(\log_{10}N) + 2.54(\log_{10}N)^2$	$27.75 - 8.60(\log_{10}N) + 1.23(\log_{10}N)^2$	± 43.4
5,000-54,999 (medium)	By Equation ($25 \leq N \leq 1,000$)	$19.38 - 11.71(\log_{10}N) + 1.97(\log_{10}N)^2$	$20.64 - 6.23(\log_{10}N) + 0.97(\log_{10}N)^2$	± 33.6
55,000+ (high)	By Equation ($25 \leq N \leq 1,000$)	$26.15 - 15.59(\log_{10}N) + 2.59(\log_{10}N)^2$	$26.05 - 12.41(\log_{10}N) + 2.15(\log_{10}N)^2$	± 22.0

Table 2: Equation form of accuracy and precision limits given the sample size (N), based on historical short-term counts factored to AADT. From Table 14a on pg. 49 from Krile et al. (2024)

Next, to calculate whether we met precision benchmarks, we used Table 1 (above) and Table 3 (below) Table 1 shows the Maximum Error Range for each sample data point, and Table 3 shows the maximum number of times (c) that site errors can exceed the Maximum Error Range and still meet the precision benchmark for a particular sample size (column n).

Sample Size (n)	Maximum Number of Sample Sites Exceeding Maximum Error Range (c)
25	0
50	1
80	2
100	4
200	7
400	16
1,000	50

Table 3: Maximum number of samples sites that can exceed the maximum error rate (found in Table 2 above based on volume category) by sample size. From Table 15 on pg. 54 from Krile et al. (2024)

Cross-Validation Results

We evaluate accuracy using stratified k-fold cross-validation (CV). In this technique, a subset of the ground-truth data is held out, the model is trained on the remaining data, and error metrics are calculated using the holdout group. We used “k” equals 10, or 10 folds, which is industry standard. To better estimate the model’s ability to generalize on unseen sites, we stratified by road volume category and implemented distance-based rules to make sure that counters in close proximity were grouped together in training and testing the CV fold. The results represent the expected error at a location on which the model has not been trained.

Table 4 shows the cross-validation results for the volume model for all counters with more than 10 months of 2024 data, broken down by road size category. Overall, these results meet

expectations and are within range of AADT estimates from previous years. Notably, the results show that bias is low—lower than $\pm 2\%$ on all but the smallest roads. MAPE is under 20% for all size categories and less than 10% for all size categories above 5,000 AADT. The 68th percentile absolute errors range from 22.5% for very low volume roads to under 7.5% for larger roads. For example, road size G has a 68th percentile error of 7.1%, indicating that the volume error for a typical road with about 45,000 in daily traffic would be about 7%, or 3,200 as an absolute number.

In general, errors decrease with larger road size, which is expected with errors measured in percentage terms, since higher volume roads have a larger denominator.

Road Size (AADT)	Number of Counters	Bias (%)	68th Percentile Absolute Error (%)	95th Percentile Absolute Error (%)	MAPE (%)	NRMSE (%)
A: 0 – 499	63	5.9	22.5	54.4	18.1	25.9
B: 500 – 1,999	364	1.6	14.8	30.9	12.1	16.5
C: 2,000 – 4,999	589	-1.3	13.4	27.1	10.9	14.6
D: 5,000 – 9,999	672	-1.3	11.8	23.5	9.4	12.1
E: 10,000 – 19,999	789	-1.3	9.3	20.0	7.8	10.1
F: 20,000 – 34,999	686	-0.8	8.3	20.1	7.3	10.1
G: 35,000 – 54,999	592	-1.3	7.1	19.1	6.7	9.4
H: 55,000 – 84,999	607	-1.6	6.7	16.6	6.0	8.2
I: 85,000 – 124,999	378	-1.9	7.2	20.2	6.4	9.3
J: 125,000+	353	-2.5	7.2	16.6	6.2	9.1

Table 4: Cross-validation results for StreetLight AADT 2024 by road size

Predictive Power

StreetLight 2024 AADT estimates strongly correlate with ground-truth AADT, as shown in Figure 2. The best-fit line's r-squared is 0.987.

Correlation between Counter AADT and Estimated AADT

R-squared = 0.987

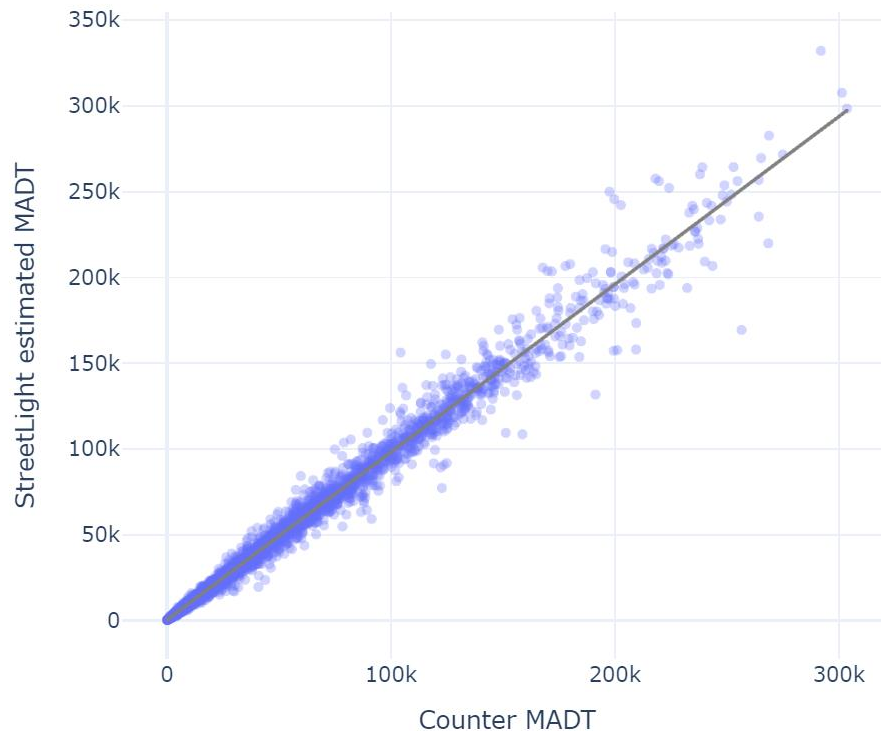


Figure 2: Estimated AADT 2024 values correlate very closely with those reported by permanent counters.

Comparison to Updated FHWA Benchmarks

We compared the accuracy and precision of StreetLight's AADT 2024 model estimates to the FHWA target benchmarks in Table 5, below. The table includes the target benchmarks and results from StreetLight's AADT model.

- Accuracy (Bias and MAPE): We used the formulas for accuracy in Table 2 to calculate targets based on the number of counter sites we created estimates for. We called this sample size.
- Precision (number of sites with errors beyond target): We determined the maximum number of sites beyond the target error range by extrapolating between sample size levels of the maximum number of sample sites exceeding maximum error range from Table 3.

The cell shading highlights how the model estimates perform:

- The green shading highlights where StreetLight's AADT estimate performed equal to or better than the target benchmark.
- The yellow shading indicates where StreetLight's AADT was within two percentage points of the target benchmark

- Since median bias can be either positive or negative, we compare the absolute values of the errors.

AADT Volume Category	Sample Size	Bias (%)		MAPE (%)		Number of Sites Beyond Target Absolute Error Range (Number of Outliers)	
		Model Results	Target	Model Results	Target	Model Results	Target (Max sites beyond target)
500-4,999	341	-1.46	± 2.33	11.09	13.86	4 sites >43.4% error	11 sites >43.4% error
5,000-54,999	1,254	-3.09	± 2.01	8.14	10.47	6 sites >33.6% error	50 sites >33.6% error
55,000+	830	-2.20	± 2.71	6.18	8.14	23 sites >22% error	40 sites >22.0% error

Table 5: Cross-validation results and calculated targets for StreetLight AADT 2024 by road size, shaded to compare with short-term count expansion errors.

The results show that StreetLight's AADT model almost completely outperforms the new benchmarks.

- StreetLight performed nearly two percentage points better than the target MAPE across all three volume categories. **This means that user can expect model AADT estimates to be, on average, closer to actual counter AADT volumes than 48-hour temporary traffic volume studies.**
- We had fewer outliers than permitted in the benchmark. For example, for a sample size of 341 in the 500-4,999 volume category, 11 sites with absolute errors beyond 43.4% were allowed. In that category only 4 sites had absolute errors above 43.4%.
- In one category, medium-volume trips, bias was beyond the benchmark range by one percentage point. Bias is negative in all categories, meaning that the StreetLight model AADT estimates tend to be slightly below counter AADTs.

Evaluating AADT Results Across Time

StreetLight provides AADT estimates from 2019 through 2024. These estimates can be compared across time on a given road to assess annual change on a roadway or applied to use cases like before-and-after studies.

A validation of our annual time trends compared to permanent counters suggests that our reported time trends are strong, **especially for medium to higher volume roads with an expected year-over-year percent change between 5% and 25%.** Smaller year-over-year

changes may be below our threshold of detection. Fluctuations of 25% or more are rare and often associated with lower volume roads (for example a road that saw 100 AADT in 2023 and 130 in 2024 will show a fluctuation of 30%).

As with many of our metrics, lower volume roads, particularly those under 500 AADT, are prone to higher error and thus the ability to detect year-over-year time trends reported at these locations will be less accurate.

Validation

To validate our ability to detect year-over-year changes, we relied on 3,520 permanent counters across 20 states. In this case, each location could have counters capturing each direction of travel on a roadway. All permanent counters used in this validation were required to have AADT reported across all calendar years from 2019-2024. For each location we calculated the year-over-year percent change as reported by the permanent counters and StreetLight's AADT estimates. For example, a counter that reported AADT of 5,000 one year, and 5,500 the next year, would have a 10% year-over-year change.

In Figure 3a below we visualize the distribution of year-over-year change as reported by StreetLight and permanent counters for six consecutive years from 2019 through 2024. The histogram shows that distributions are centered around zero, with very few instances of changes greater than 25%. The changes that occurred from 2019-2021 due to the impacts of the COVID-19 pandemic leading to a higher proportion of roads with year-over-year changes in the positive and negative 5-25% ranges. From 2019-2020 many roads experienced a notable decrease in traffic, hence the higher number of counters in that range in Figure 3a. While from 2020 to 2021 many roads experienced a notable increase in traffic to correct for some of the loss that occurred in 2020.

Distribution of Year Over Year Change 2019-2024

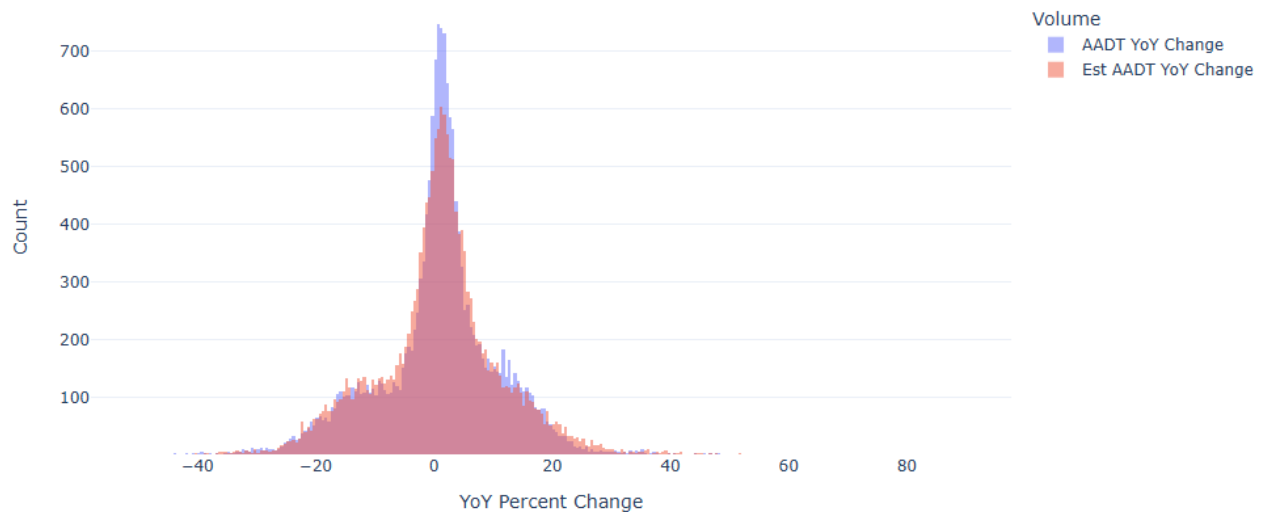


Figure 3a: StreetLight's year-over-year changes in AADT estimates (in red) share a very similar distribution to those reported by permanent counters (in blue) from 2019-2024. Year-over-year changes in the positive and negative 5% - 25% range are higher than expected due to steep increases and declines in traffic that occurred during the COVID-19 pandemic.

In Figure 3b we exclude the 2019-2020 and 2020-2021 year-over-year change statistics from the visualization, you can see that the distribution is normal and tightly clustered around zero. There appears to be a slight skew towards positive year-over-year change, which would make sense given a steady recovery from the COVID-19 pandemic. This suggests that under normal conditions, year-over-year change tends to be minimal.

Distribution of Year Over Year Change 2021-2024

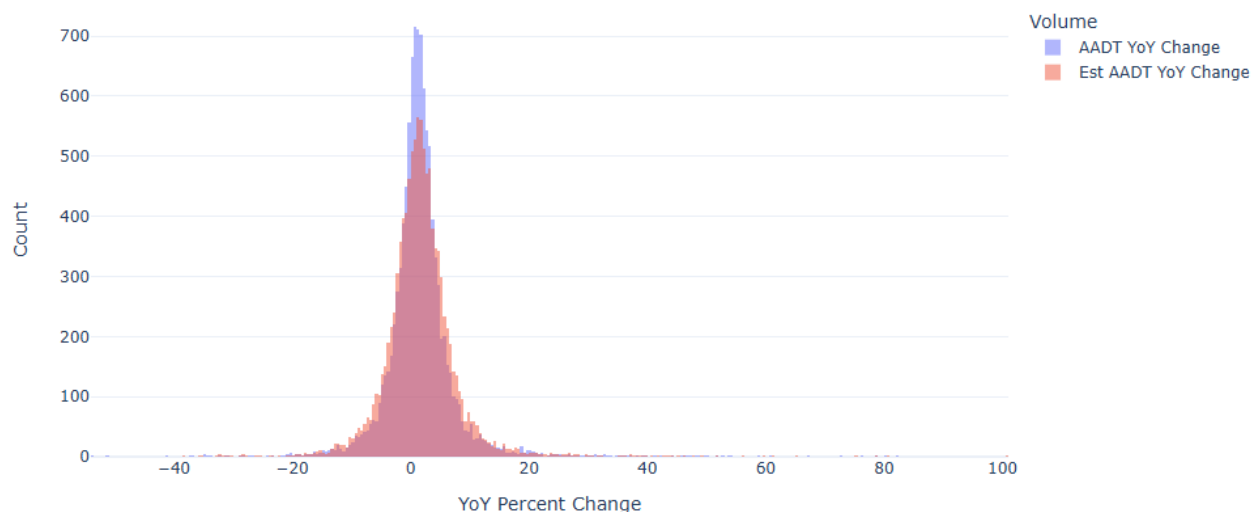


Figure 3b: StreetLight's year-over-year changes in AADT estimates (in red) share a very similar distribution to those reported by permanent counters (in blue) from 2021-2024. Year-over-year changes are tightly clustered around zero.

Overall, StreetLight's estimates produce a very similar distribution pattern to those reported by the permanent counters. StreetLight can capture the large and moderate swings in traffic caused by the COVID-19 pandemic, as well as the standard year-over-year variation that occurs in the proceeding years. StreetLight's AADT estimates reflect a slightly wider distribution relative to permanent counters due to model error.

In addition to evaluating distributions, we assessed the correlation between year-over-year percent change as reported by permanent counters vs. StreetLight's AADT estimates.

Across all road sizes and counters, correlation is 0.77, indicating a strong relationship between the two variables. It's important to consider road size when evaluating the quality of AADT estimates. Small swings in traffic on a low-volume road can present as large percent changes. Figure 4 below depicts the correlation between year-over-year percent change values color-coded by road size.

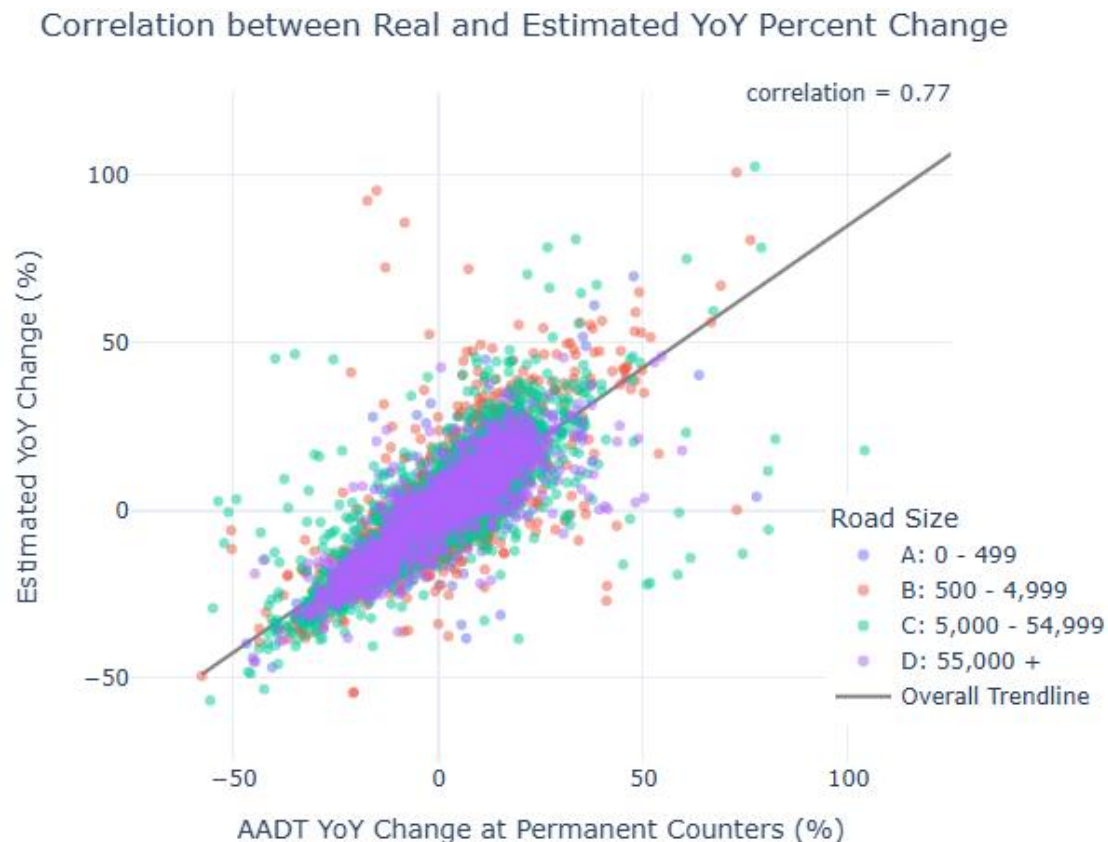


Figure 4: Correlation between year-over year change reported by permanent counters vs. StreetLight AADT estimates. Correlation is 0.77 when all four road classes are included.

Figure 4 shows that outliers tend to correspond with lower volume roads, while larger volume roads (Category D in purple) are clustered most tightly around the overall trendline. This is also reflected when the correlation is broken down by these four road classes in Table 7.

AADT Volume Category	Number of Counters	Correlation
A: 0-499	312	0.38
B: 500-4,999	4,492	0.62
C: 5,000-54,999	11,238	0.84
D: 55,000+	5,078	0.89

Table 7: Correlation between year-over year change reported by permanent counters vs. StreetLight AADT estimates broken down by four road classes. Correlation is very strong on the highest volume roads and weaker on the lowest volume roads.

The high correlation on higher volume roads with 5,000 AADT and above confirms that we're able to detect year-over-year change well in many circumstances. The lower correlation on the lowest volume category, coupled with a smaller number of counters, suggests that we can expect more variability when evaluating year-over-year change on very low volume roads.

Figures 5a and 5b below illustrate this concept on two sample roads. Where the time series chart in Figure 5a depicts variation that could occur on a lower volume road, while the time series chart in Figure 5b depicts more consistent results associated with a larger volume road. On the lower volume road, StreetLight AADT is within range of AADT reported by the permanent counter, however the year-over-year percent change will be more variable.

AADT Values at Select Permanent Counters: Ohio 55157 (A: 0 - 499)

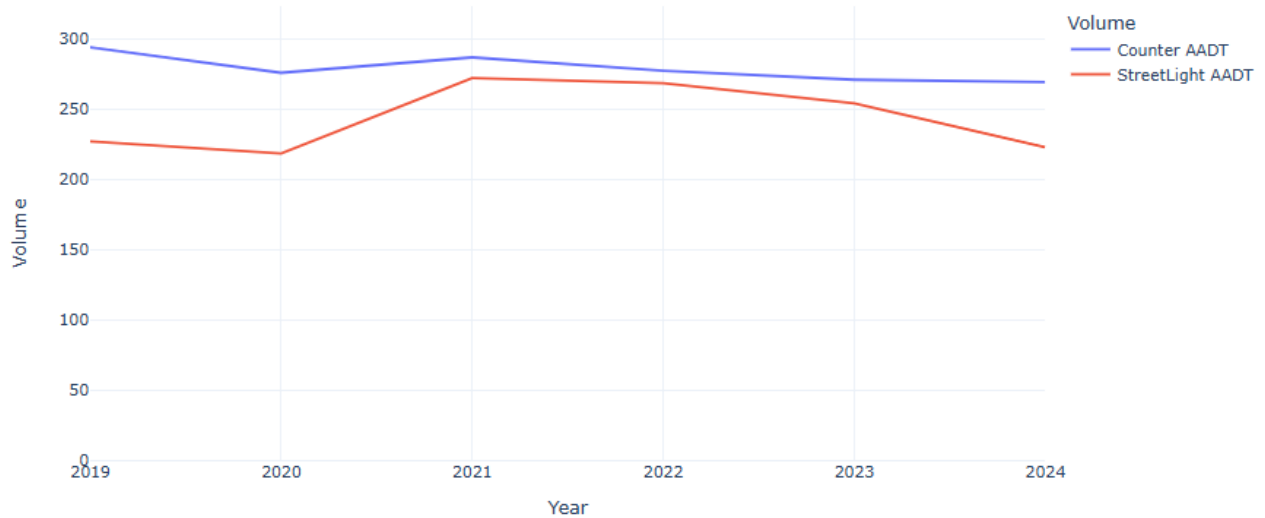


Figure 5a: Counter 55157 in Ohio (above) illustrates variation on a low volume road with values fluctuating around 250 AADT.

AADT Values at Select Permanent Counters: Colorado 000214 (F: 20,000 - 34,999)

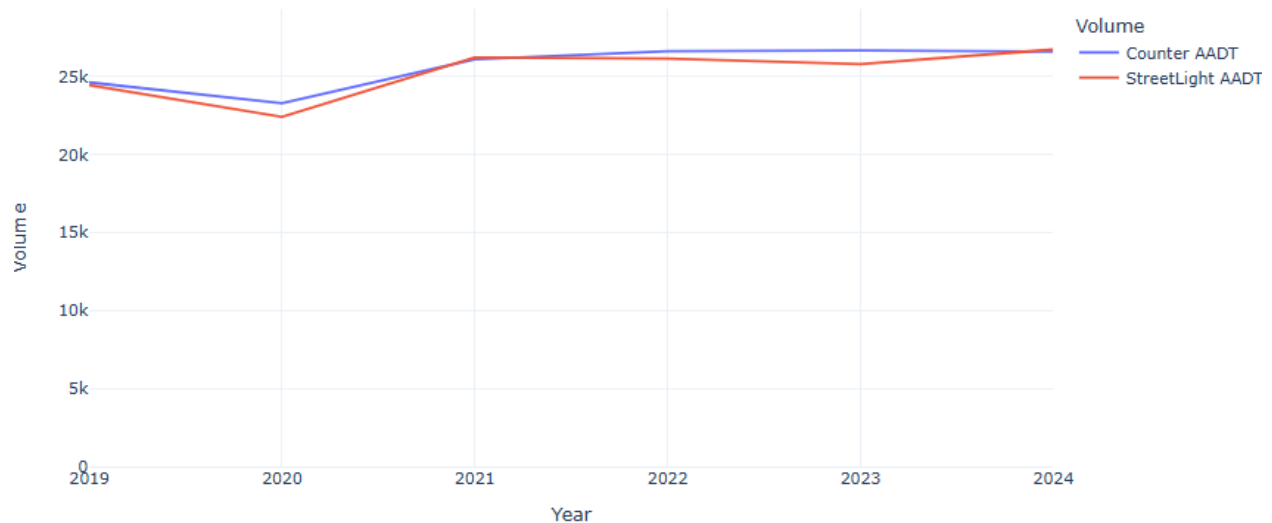


Figure 5b: Counter 000214 in Colorado (above) illustrates a higher volume road with less year-over-year variation between Counter and StreetLight AADT.

Limitations

It's important to keep in mind that our models are not built to optimize for accuracy in year-over-year percent change, instead they are built to optimize for accuracy in a particular month.

The percent change between two given years is based on the percent difference between two separate estimates, each with their own potential for error. For example, if our 2023 AADT estimate for a given location has a negative bias, while our 2024 AADT estimate for the same location has a positive bias, the year-over-year time trend might reflect a larger percent change between 2023 and 2024, whereas in reality, the actual change may have been minimal. In the future, we'll work to optimize predictions to more precisely capture monthly and annual time trends. Additionally, users can work with our support and solutions teams if they have questions about particular time trend analyses.

About StreetLight Data

StreetLight Data, Inc. ("StreetLight") pioneered the use of Big Data analytics to shed light on how people, goods, and services move, empowering smarter, data-driven transportation decisions. StreetLight's proprietary data processing engine, Route Science®, algorithmically transforms its vast data resources to measure travel patterns of vehicles, bicycles, and pedestrians, accessible as analytics on the StreetLight InSight® SaaS platform. Acquired by Jacobs as a subsidiary in February 2022, StreetLight provides innovative digital solutions to help communities reduce congestion, improve safe and equitable transportation, and maximize the positive impact of infrastructure investment.



STREETLIGHT

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